Waste Classification using CNN and Deep Learning for Sustainability

Rapid urbanization and industrial growth have significantly increased global waste generation, posing major environmental and health concerns. Improper segregation of waste leads to soil contamination, water pollution, and resource loss. As societies move toward sustainable living, the need for intelligent waste management systems has become more urgent than ever. Technologies like Artificial Intelligence (AI) and Deep Learning offer an opportunity to address this issue efficiently by automating waste identification and segregation.

Project Problem Statement

This project focuses on developing an intelligent system based on **Convolutional Neural Networks (CNNs)** and Deep Learning to classify waste materials from images accurately. The system aims to identify and categorize waste into types such as plastic, glass, paper, metal, and organic matter. By integrating CNN-based computer vision techniques, the project seeks to automate the segregation process, reducing human intervention and errors. The model will be trained on publicly available datasets from **Kaggle** and further refined using **Google Teachable Machine** to enhance recognition accuracy and deployment flexibility.

The ultimate goal is to contribute to **sustainable waste management practices** and reduce the environmental impact caused by improper disposal. This project aligns with the **United Nations Sustainable Development Goals (SDG 12 – Responsible Consumption and Production)** by promoting recycling and reducing landfill waste through intelligent automation.

Dataset

Dataset Name: Garbage Classification Dataset (Version 2)
Source: Kaggle – A comprehensive image dataset for garbage classification and recycling research.

This dataset consists of thousands of labeled images categorized into different classes such as cardboard, glass, metal, paper, plastic, and trash. It is designed for machine learning and computer vision applications related to recycling and waste management. The dataset is ideal for developing CNN-based image classification models to support sustainability-driven research.

Next Steps

- 1. Collect and Prepare Dataset from Kaggle.
- 2. Preprocess images (resize, normalize, and organize by class).

- 3. Train the CNN Model using Google Teachable Machine.
- 4. Evaluate the Model Performance using accuracy and confusion matrix.
- 5. Build a Simple Web Interface using Python Streamlit for real-time testing.
- 6. Test the Complete Application with real-world waste images.
- 7. Deploy and Document the System for presentation or demonstration.

This project serves as a practical implementation of AI for environmental sustainability. By combining Deep Learning with real-world datasets, students and researchers can gain hands-on experience in building smart solutions for one of the most pressing issues of modern society — waste management. Through innovation and technology, this initiative aims to inspire eco-conscious practices for a cleaner and greener future.